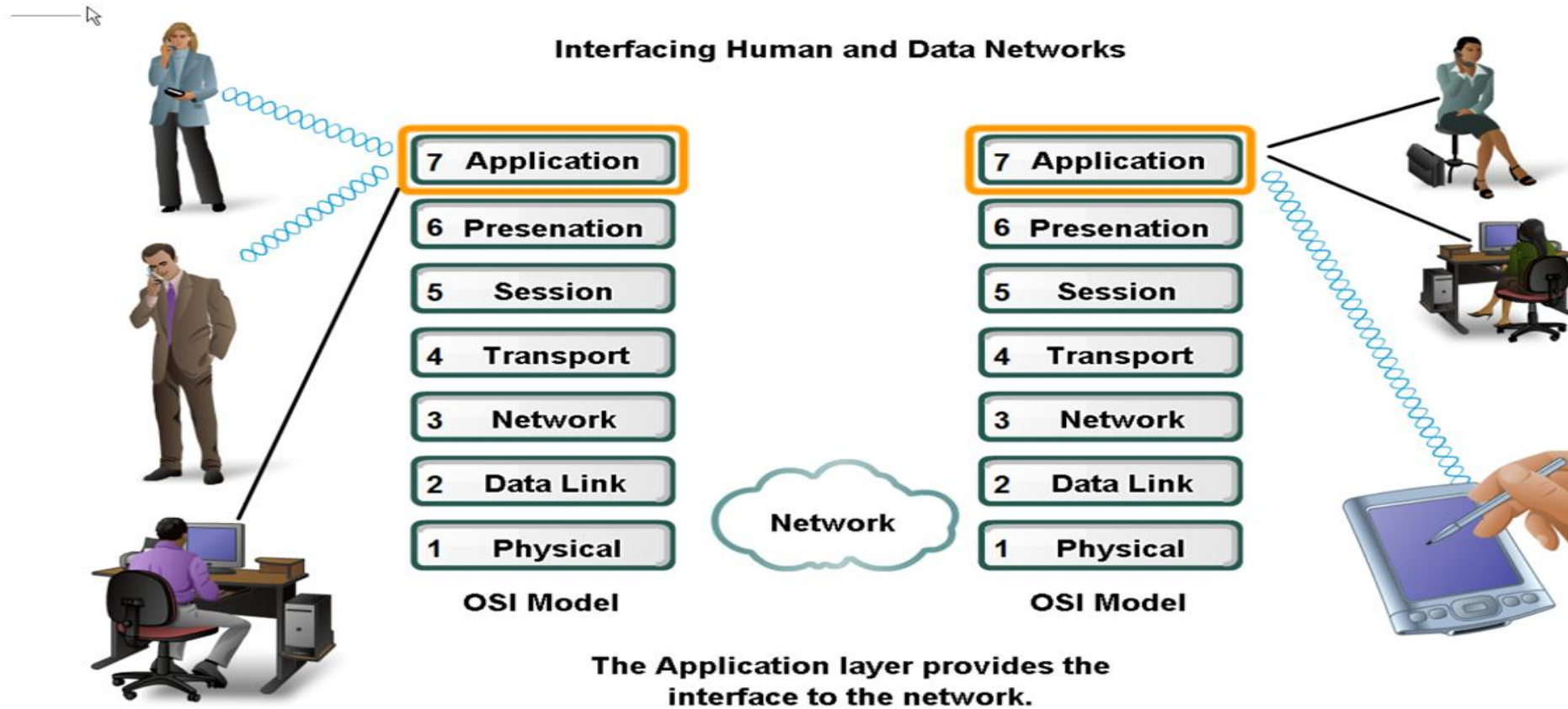


APPLICATION LAYER PROTOCOLS

Applications Layer – allows user to interface with the network!



What Is the Application Layer

The application layer provides the methods and rules that applications use when sending and receiving information.

It does not handle the physical delivery of data. That responsibility falls to the layers beneath it.

The application layer focuses on presenting data in a meaningful format for both the application and the user.

Application Protocols:

At Layer 7, we encounter various application-level protocols such as HTTP (Hypertext Transfer Protocol) for web browsing, SMTP (Simple Mail Transfer Protocol) for email, FTP (File Transfer Protocol) for file sharing, and many others.

These protocols enable different applications on different devices to communicate over networks.

User Interfaces:

Think of the applications you use daily — web browsers, email clients, instant messaging apps, and more.

These operate at Layer 7.

It's the layer where you interact with software, making it the most tangible layer for end-users.

Data Presentation:

Layer 7 handles **data translation, encryption, and compression.**

It ensures that data is presented in a format that the receiving application can understand.

Application Services:

Authentication, Authorization and Accounting (AAA) are vital services provided by Layer 7.

They are essential for secure and controlled access to network resources, protecting sensitive information.

APIs :

For developers, Layer 7 is a goldmine.

It provides APIs that allow the creation of applications using network services and protocols.

These APIs empower software to communicate effectively over networks.

Message Formatting:

Data exchanged at Layer 7 is typically in the form of messages or data packets, each with specific structures and formats understood by the receiving application.

It ensures that data arrives intact and is usable.

Security:

In the age of cybersecurity threats, Layer 7 plays a pivotal role in application-level security.

Firewalls, access control lists, and application-layer filtering are implemented at this layer to protect against various threats.

At the centre of all of this is the Transmission Control Protocol / Internet Protocol (TCP/IP), which plays a crucial role in how data moves across networks.

But **TCP/IP** is just the beginning.

In this class we are going to break down the most important protocols, **TCP, UDP, DNS, FTP, SMTP, and HTTP**

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TCP/IP

Transmission Control Protocol / Internet Protocol is a set of standardized rules allowing computers to communicate on a network like the Internet.

It ensures messages get sent correctly, like putting addresses on letters and using routers to deliver them.

The TCP/IP protocols suite consists of several protocols that work together to enable communication across networks. some of the important ones are:

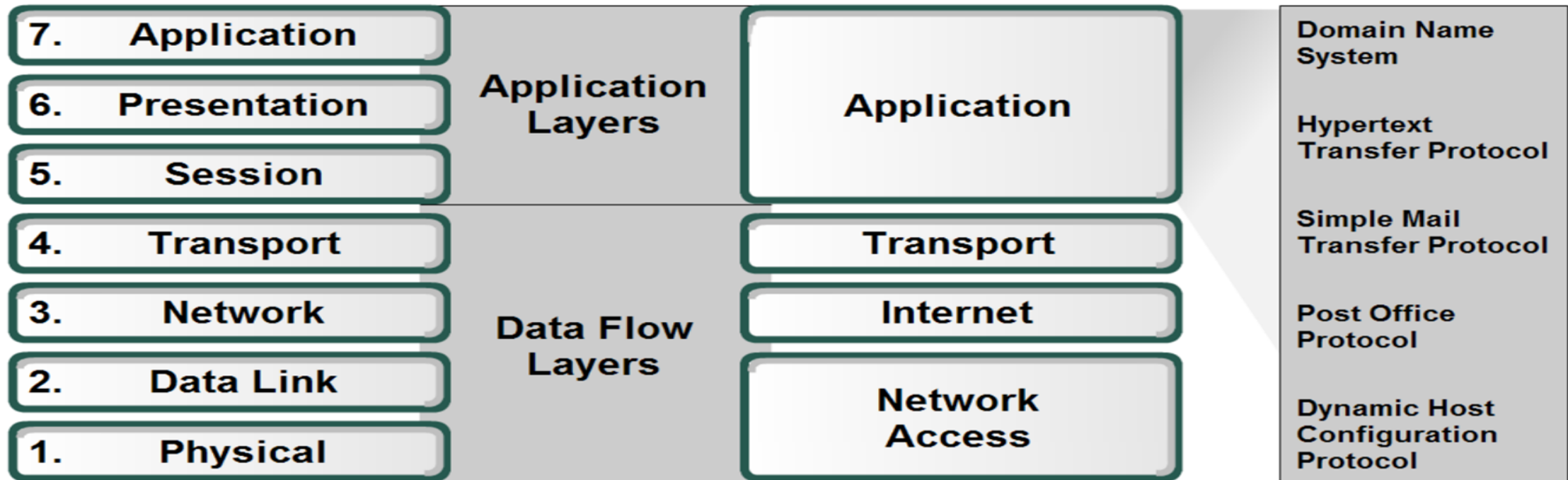
-
1. Internet Protocol (IP)
 2. Transmission Control Protocol (TCP)
 3. User Datagram Protocol (UDP)
 4. Domain Name System (DNS)
 5. File Transfer Protocol (FTP)
 6. Simple Mail Transfer Protocol (SMTP)
 7. Hypertext Transfer Protocol (HTTP)

Application Layer – Provides the interface between the applications on either end of the network.



OSI Model

TCP/IP Model



Application layer software

2 types

- Applications – Provide the human (user) interface. Relies on lower layers to complete the communication process.
-
- Services – establish an interface to the network where protocols provide the rules and formats that govern how data is treated..

How data requests occur & are filled

Client/server model

- Advantages:
- Centralized administration
- Security is easier to enforce

Application layer services and protocols

Peer-to-peer networking and applications

Client/Server Model

Client –

- device requesting information (initiates the data exchange)
 - Can also UPLOAD data to the servers
-

Server – device responding to the request

- How does it handle multiple request from multiple users and keep everything in order?
- Relies on support from the lower layer functions to distinguish between services and conversations.
- Server relies on a service called a server daemon – runs in the background and ‘listens’ for requests for that service. It can then exchange messages as appropriate & send requested data.

Examples:

- E-mail Client on an employee computer issues a request to the e-mail server for any unread e-mail. The server responds by sending the e-mail to the client.
- Conversations can originate with either party.

Peer-to-Peer (P2P) Network Model

Two or more computers are connected and are able to share resources **without having a dedicated server**

Every end device can function as a client or server on a 'per request' basis

Resources are **decentralized** (information can be located anywhere)

Difficult to enforce security and policies

User accounts and access rights have to be set individually on each peer device

Common Port Numbers

TCP

- FTP – 20-21
- Telnet – 23
- SMTP – 25
- DNS – 53 (Both TCP & UDP)
- HTTP – 80

UDP

- DHCP – 67 & 68
- POP – 110

Internet Protocol (IP)

its major function is to provide the addressing and routing mechanisms necessary for the delivery of data packets across networks.

it's just like phone numbers for computers.

When you want to send a message or a picture to someone on the internet, your computer breaks it into smaller pieces called packets. Each packet is like a little piece of the message. It puts the IP address of the person you want to send it to on each packet, like putting their address on an envelope.

Go through different routers, which are like post offices

Transmission Control Protocol (TCP)

a communication standard that enables application programs and computing devices to exchange messages over a network.

It is designed to send packets across the internet and ensure the successful delivery of data and messages over networks.

You have some packages with different labels and patterns and you want to get them delivered to a friend in another city, TCP is that magical system that helps make sure all the packages get to your friend safely and in the right order.

User Datagram Protocol (UDP)

a communication protocol that is used for time-critical data transmission, it's a connectionless protocol that offers a simpler, faster, and less reliable delivery mechanism compared to TCP,

It is often used for applications that can tolerate some degree of data loss, such as real-time audio and video playback, online gaming, etc.

It sends small packets of information, like little balls, to your friend's computer. It doesn't wait for a response or confirmation. It just keeps sending more and more packets quickly. If some packets get lost or don't make it to your friend's computer, UDP doesn't worry too much.

DNS

The **Domain Name System (DNS)** is a central part of the Internet, providing a way to match names (a website you're seeking) to numbers (the address for the website).

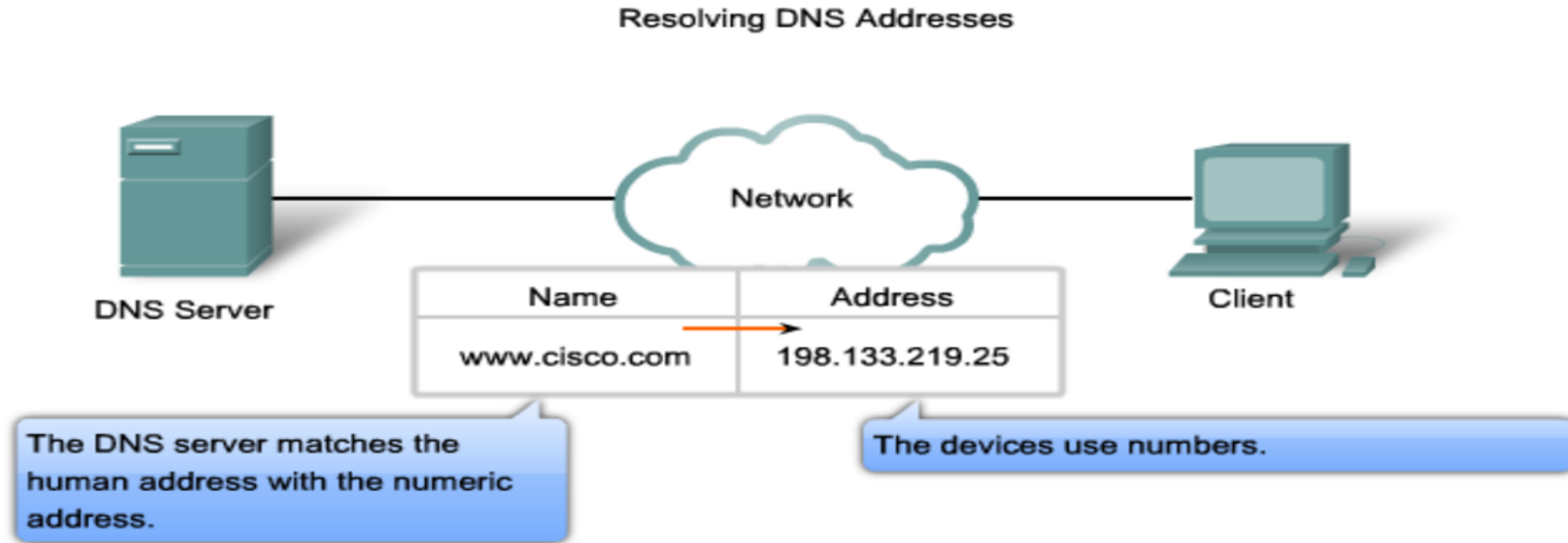
Anything connected to the Internet - laptops, tablets, mobile phones, websites - has an Internet Protocol (IP) address made up of numbers.

Your favourite website might have an IP address like 64.202.189.170, but this is obviously not easy to remember.

However a domain name such as bestdomainnameever.com is something people can recognize and remember.

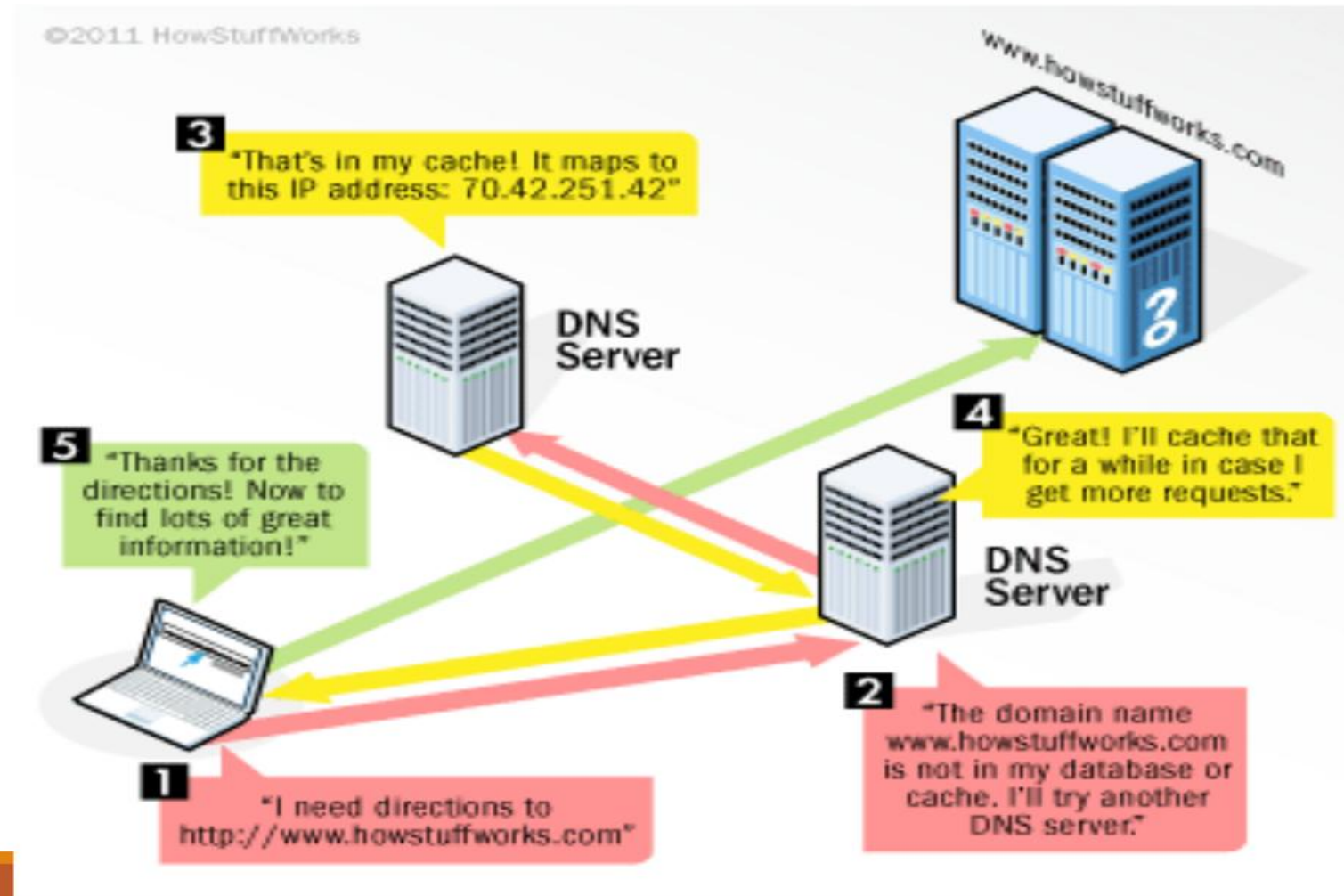
DNS syncs up domain names with IP addresses enabling humans to use memorable domain names while computers on the Internet can use IP addresses.

DNS Services and Protocol



DNS Servers resolve names to IP addresses. It would be difficult to remember the IP address of every website we like to visit, but we can remember names.

When you enter a URL into your Web browser, your DNS server uses its resources to resolve the name into the IP address for the appropriate Web server.



Domain Name System Architecture

The Domain name system comprises of **Domain Names, Domain Name Space, Name Server** that have been described below:

Domain Names

Domain Name is a symbolic string associated with an IP address. There are several domain names available;

some of them are generic such as **com, edu, gov, net** etc, while some country level domain names such as **au, in, za, us** etc.

The following table shows the **Generic Top-Level Domain names**:

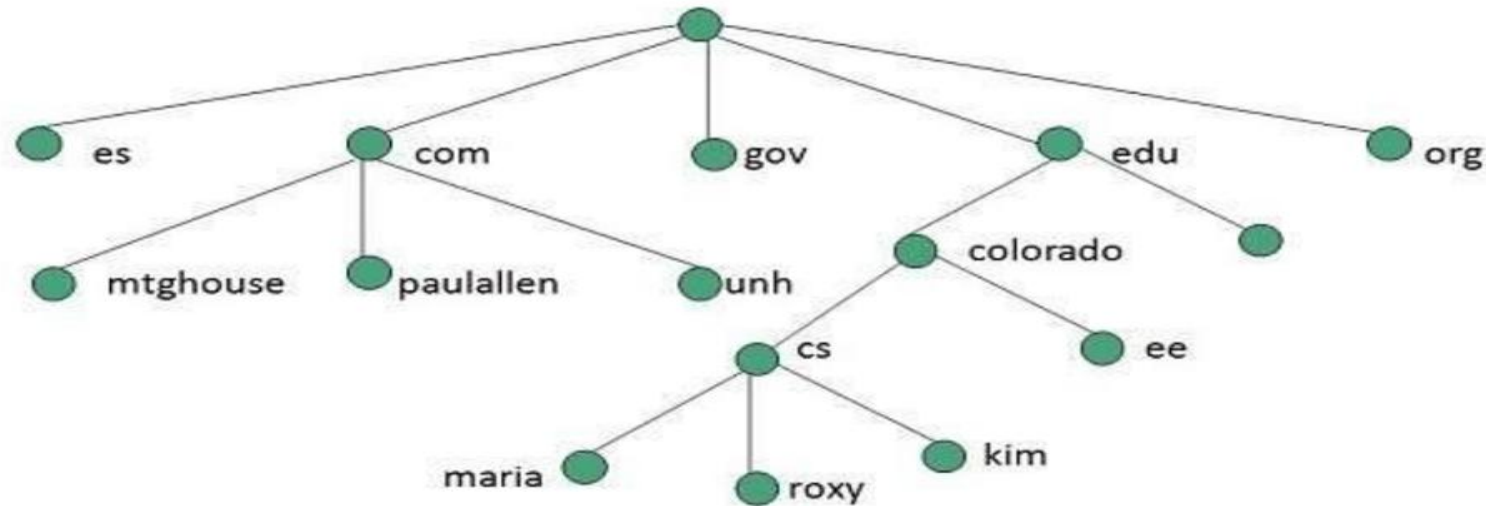
Domain Name	Meaning
Com	Commercial business
Edu	Education
Gov	U.S. government agency
Int	International entity
Mil	U.S. military
Net	Networking organization
Org	Non profit organization

The following table shows the **Country top-level** domain names:

Domain Name	Meaning
au	Australia
in	India
cl	Chile
fr	France
us	United States
za	South Africa
uk	United Kingdom
jp	Japan
es	Spain
de	Germany
ca	Canada
ee	Estonia
hk	Hong Kong

Domain Name Space

The domain name space refers a hierarchy in the internet naming structure. This hierarchy has multiple levels (from 0 to 127), with a root at the top. The following diagram shows the domain name space hierarchy:



In the above diagram each subtree represents a domain. Each domain can be partitioned into sub domains and these can be further partitioned and so on.

How Does DNS Work?



Name Server

Name server contains the DNS database. This database comprises of various names and their corresponding IP addresses.

Since it is not possible for a single server to maintain entire DNS database, therefore, the information is distributed among many DNS servers.

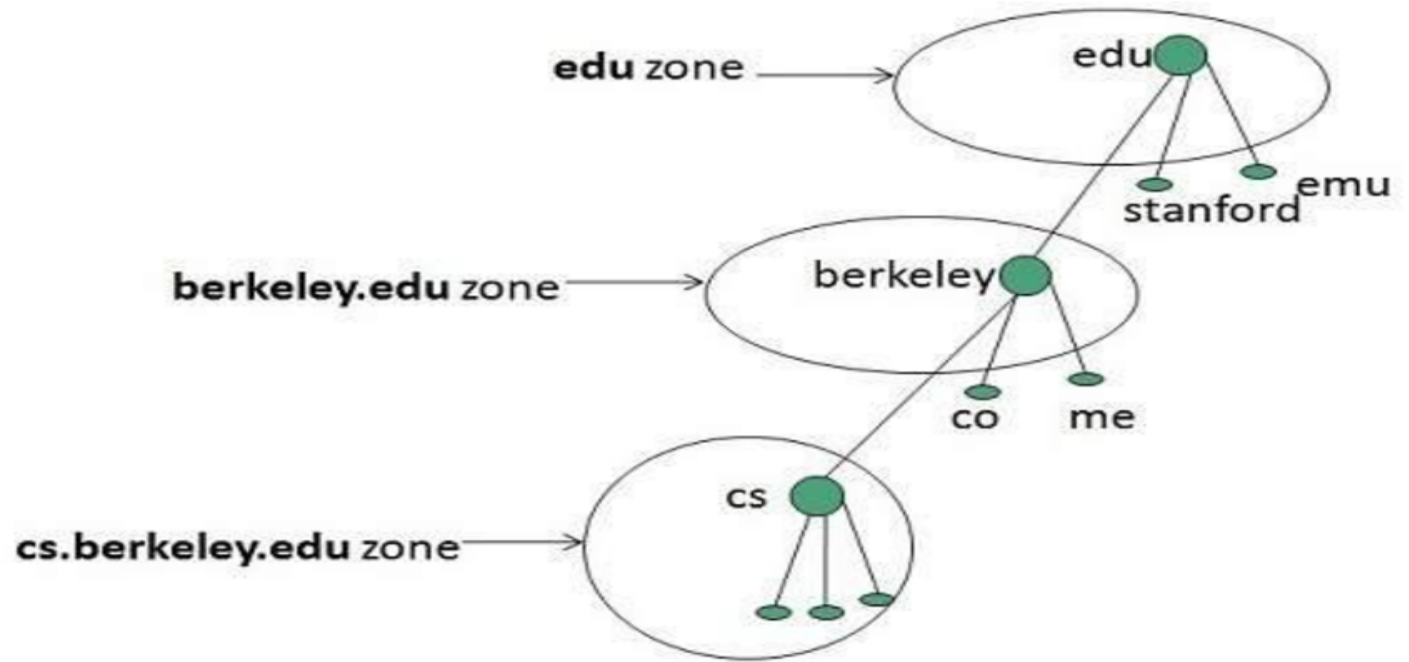
- Hierarchy of server is same as hierarchy of names.
- The entire name space is divided into the zones

Zones

Zone is collection of nodes (sub domains) under the main domain. The server maintains a database called zone file for every zone.

If the domain is not further divided into sub domains then domain and zone refers to the same thing.

The information about the nodes in the sub domain is stored in the servers at the lower levels however; the original server keeps reference to these lower levels of servers.



TYPES OF NAME SERVERS

ROOT SERVER

Root Server is the top level server which consists of the entire DNS tree. It does not contain the information about domains but delegates the authority to the other server

PRIMARY SERVERS

Primary Server stores a file about its zone. It has authority to create, maintain, and update the zone file.

SECONDARY SERVER

Secondary Server transfers complete information about a zone from another server which may be primary or secondary server. The secondary server does not have authority to create or update a zone file.

DDNS

The information contained in the domain name space must be stored. However, it is very inefficient and also unreliable to have just one computer store such a huge amount of information.

A primary server loads all information from the disk file; the secondary server

loads all information from the primary server.

When the secondary downloads information from the primary, it is called zone transfer.

Storing, locating, and transmitting information on the Web

How information is located: the URL

- ❖ To move from one page of a document to another page, or to another document on the same or another Web site, the user clicks a **hyperlink** (usually just called a link) in the document shown in their Web client.
- ❖ Documents and locations within documents are identified by an address, defined as a Uniform Resource Locator, or **URL**. The following URL illustrates the general form:

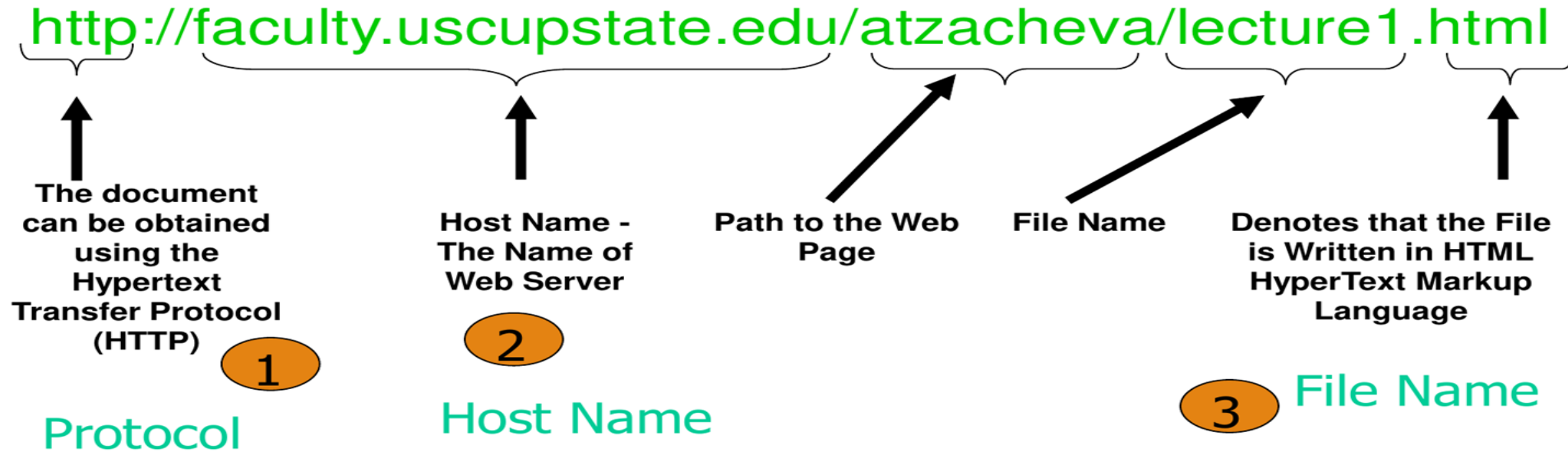
`http://www.sybase.com/products1`

or

`http://www.sybase.com/inc/corpinfo/mkcreate.html`

The Universal Resource Locator (URL)

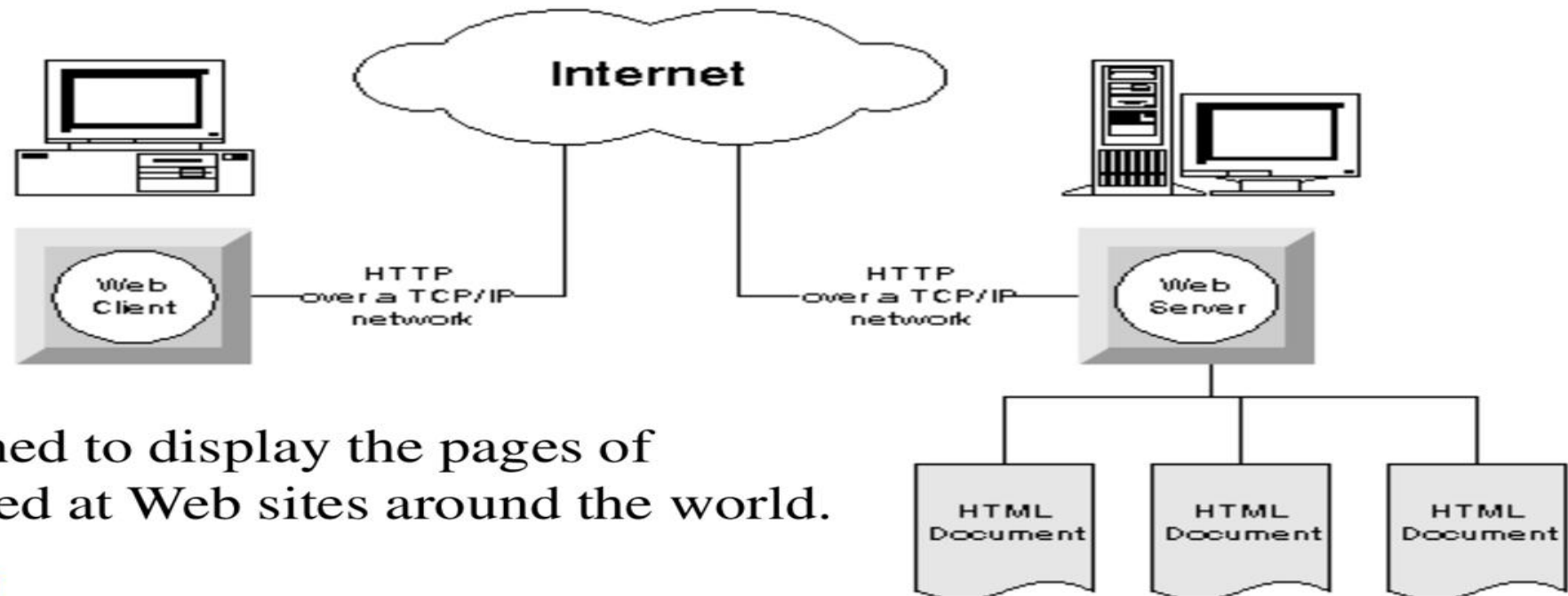
Each page of information on the web has a unique address called the URL at which it can be found.



Accessing information on the web

As Figure 1 illustrates, information is stored at **Web sites**. Access to the information is managed by a **Web server** for the site. Users access the information using **Web clients**, which are also called **browsers**.

Figure 1: Accessing information on the Web



Browser is designed to display the pages of information located at Web sites around the world.

Accessing information on the web

Information on the Web is stored in documents, using a language called **HTML** (HyperText Markup Language).

Web clients must interpret HTML to be able to display the documents to a user.

The protocol that governs the exchange of information between the Web server and Web client is named **HTTP** (HyperText Transfer Protocol).

About HTML

Programming Language

Used to create web pages

Use Hypertext documents for use on WWW.

About HTTP

The standard Web transfer protocol is **HTTP (HyperText Transfer Protocols)**.

It is used for document exchange between servers and clients (typically browsers) in the WWW.

To retrieve a document, the client first sends a request to the web server and waits for a reply

An HTTP service (a program that waits for http requests) on the server then handles the request and the document is sent to the client (over a connection established using TCP/IP – Transmission Control Protocol / Internet Protocol)

HTTP

1. HTTP transfer the browser sends a request for a document to the server. Included in this request is the description of the document being requested, as well as a list of document types that the browser is capable of handling.

The Multipurpose Internet Mail Extensions (MIME) standard is used to specify the document types that the browser can handle. The browser is able to specify weights for each document type, in order to inform the server about the relative desirability of different document types.

2. In response to a query the server returns the document to the browser using one of the formats acceptable to the browser.

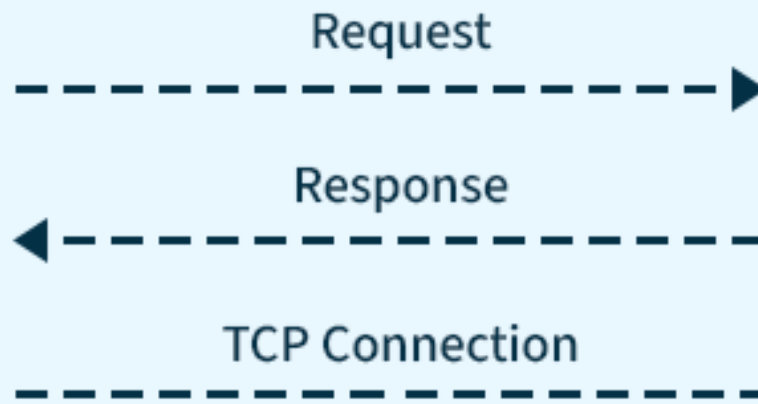
If necessary the server can translate the document from the format it is stored in into a format acceptable to the browser.



HTTP Connection



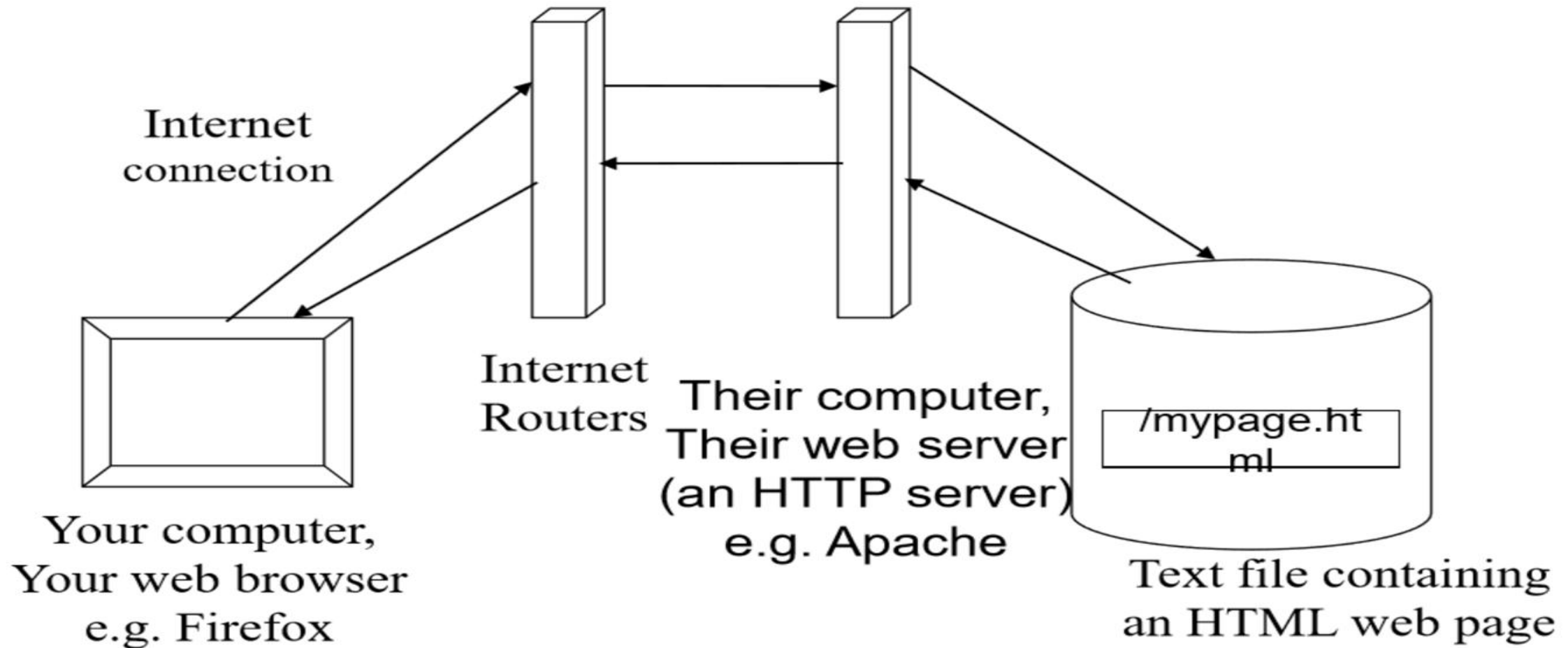
Client



Server

How does a Web Browser (Firefox) fit in this picture?

- your browser connects, using the HTTP protocol, to a web server
 - the web server fetches an HTML web page and sends the HTML to your browser
 - your browser turns the HTML page into a nice-looking page on your screen



Understanding HTTP Request and Response

An HTTP request is how your browser asks the server for something. It includes:

- **HTTP Version:** The version of HTTP (like HTTP/1.1 or HTTP/2) being used.
- **URL:** The specific address of the resource (e.g., <https://www.example.com/about>).
- **HTTP Method:** The type of action being requested (e.g., GET to retrieve information or POST to send data).
- **HTTP Request Headers:** Extra information about the request, like what kind of browser you're using or what kind of content you're expecting.
- **HTTP Request Body:** In some cases, the request will include a body that contains data (e.g., when you submit a form).

2. HTTP Response

An HTTP response is the server's answer to your request. It includes:

- **HTTP Status Code:** A number that tells you if the request was successful or not (e.g., 200 OK means everything is fine, 404 Not Found means the requested page doesn't exist).
- **Response Headers:** Information about the response, like what kind of data is being sent (e.g., Content-Type: text/html means it's an HTML page).
- **Response Body:** The content that the server sends back (e.g., HTML code that the browser will use to display the webpage).

1xx – Informational

- Indicates that the request has been received and is being processed.
- No final response yet; client should continue.

Example: 100 Continue

2xx – Success

- The request was successfully received, understood, and processed.
- Confirms that the operation was completed correctly.

Examples:

- *200 OK – Request successful*
- *201 Created – New resource created*

3xx – Redirection

- Additional action required to complete the request.
- The client is redirected to another URL.

Examples:

- *301 Moved Permanently – Resource permanently shifted*
- *302 Found – Temporary redirection*

4xx – Client Error

- Errors caused by invalid requests from the client side.
- Needs correction before retrying.

Examples:

- *400 Bad Request – Invalid syntax*
- *401 Unauthorized – Authentication required*
- *404 Not Found – Resource not available*

5xx – Server Error

- The server fails to fulfill a valid request.
- “The issue lies on the server side.

Examples:

- *500 Internal Server Error – Generic server failure*
- *503 Service Unavailable – Server overloaded or down*

History of HTTP

Tim Berners-Lee and his team at CERN get credit for inventing the original HTTP and associated technologies.

- **HTTP version 0.9:** This was the first version of HTTP, which was introduced in 1991.
- **HTTP version 1.0:** In 1996, RFC 1945 (Request For Comments) was introduced in HTTP version 1.0.
- **HTTP version 1.1:** In January 1997, RFC 2068 was introduced in HTTP version 1.1. Improvements and updates to the HTTP version 1.1 standard were released under RFC 2616 in June 1999.
- **HTTP version 2.0:** The HTTP version 2.0 specification was published as RFC 7540 on May 14, 2015.
- **HTTP version 3.0:** HTTP version 3.0 is based on the previous RFC draft. It is renamed as Hyper-Text Transfer Protocol QUIC which is a transport layer network protocol developed by Google.

Comparing HTTP, HTTP/2, and HTTP/3

HTTP/2: Improved Performance

HTTP/2 is an improved version of HTTP introduced in 2015. It made several changes to make websites load faster:

- **Multiplexing:** Multiple requests can be sent over one connection at the same time, reducing delays.
- **Header Compression:** HTTP/2 compresses the data sent in headers to make it smaller and faster.
- **Server Push:** This allows the server to send additional resources (like images or scripts) to the browser before the browser **even asks for them**.

HTTP/3: The Latest Version

HTTP/3, released in 2022, builds on HTTP/2 but with a key improvement: it uses the QUIC protocol instead of TCP. QUIC is faster and more reliable because it:

- Reduces connection setup time.
- Handles data loss better, especially in poor network conditions.
- Offers better security by integrating encryption directly into the protocol.

Cookies in HTTP

An HTTP cookie (web cookie, browser cookie) is a little piece of data that a server transmits to a user's web browser.

When making subsequent queries, the browser may keep the cookie and transmit it back to the same server.

An HTTP cookie is typically used, for example, to maintain a user's login state and to determine whether two requests originate from the same browser. The stateless HTTP protocol, retains stateful information.

Advantages of HTTP Cookies

Authentication

Cookies can be used to verify users' identities and store their login information so they don't have to enter it each time they visit a website.

Personalization

Cookies can be used to save user settings and preferences, including language and display preferences. This enhances customers' overall user experience by enabling websites to offer them a customized experience depending on their preferences.

Management of user sessions

Cookies can be used to manage user sessions, enabling websites to monitor user behavior and retain session data like the contents of a user's shopping cart or the pages they have viewed.

For online shopping carts

Items that a user has put in their shopping cart on an e-commerce website can be stored in cookies. This enables customers to keep shopping and come back to their cart later without losing their chosen products.

Login management

Users can transition between devices without having to log in again by using cookies, which can be used to manage user logins across several devices.

HTTP cookies are a valuable tool for both website owners and users, providing a variety of benefits that aid in improving the user experience, tracking user activity, and optimizing website speed. However, its use raises some security concerns as well.

Security issues of HTTP cookies

User impersonation

The key risk with unprotected cookies is user impersonation.

Session hijacking

In session hijacking attacks, an attacker steals a user's session ID or authentication token, which can allow them to impersonate the user and gain unauthorized access to sensitive information.

Information leakage

Cookies can also store sensitive information, such as login credentials or personal data, which can be accessed by an attacker if the cookie is stolen or compromised.

Cross-site scripting (XSS) attacks

Attackers can insert malicious code into a web page using XSS attacks, which can then be used to steal user cookie data. The attacker can assume the user's identity or gain unauthorized access to confidential information if they have access to the cookie data.

HTTP Request: types of HTTP requests

There are several types of HTTP requests, each with a specific purpose and use case.

The most commonly used types of HTTP requests are

GET, POST, HEAD, PUT, DELETE, CONNECT, and OPTIONS.

Understanding the different types of HTTP requests is essential for working with HTTP and building web applications.

GET: The most commonly used HTTP request method, GET requests to retrieve data from a server. GET requests are used to retrieve a web page, images, videos, and other resources from a server. They are typically used to retrieve information and are the most common type of HTTP request.

POST: POST requests send data to a server. They are used to submit forms, upload files, and send data to a server for processing. Unlike GET requests, POST requests are not cached or stored by the browser, and the data sent with a POST request is not visible in the URL.

HEAD: Head requests are similar to GET requests, but they only retrieve the headers of a response, not the actual resource. Header requests are useful for checking the status of a resource without downloading the entire resource.

PUT: PUT requests to update an existing resource on a server or create a new resource if it doesn't exist. PUT requests are typically used to upload files or update resources on a server.

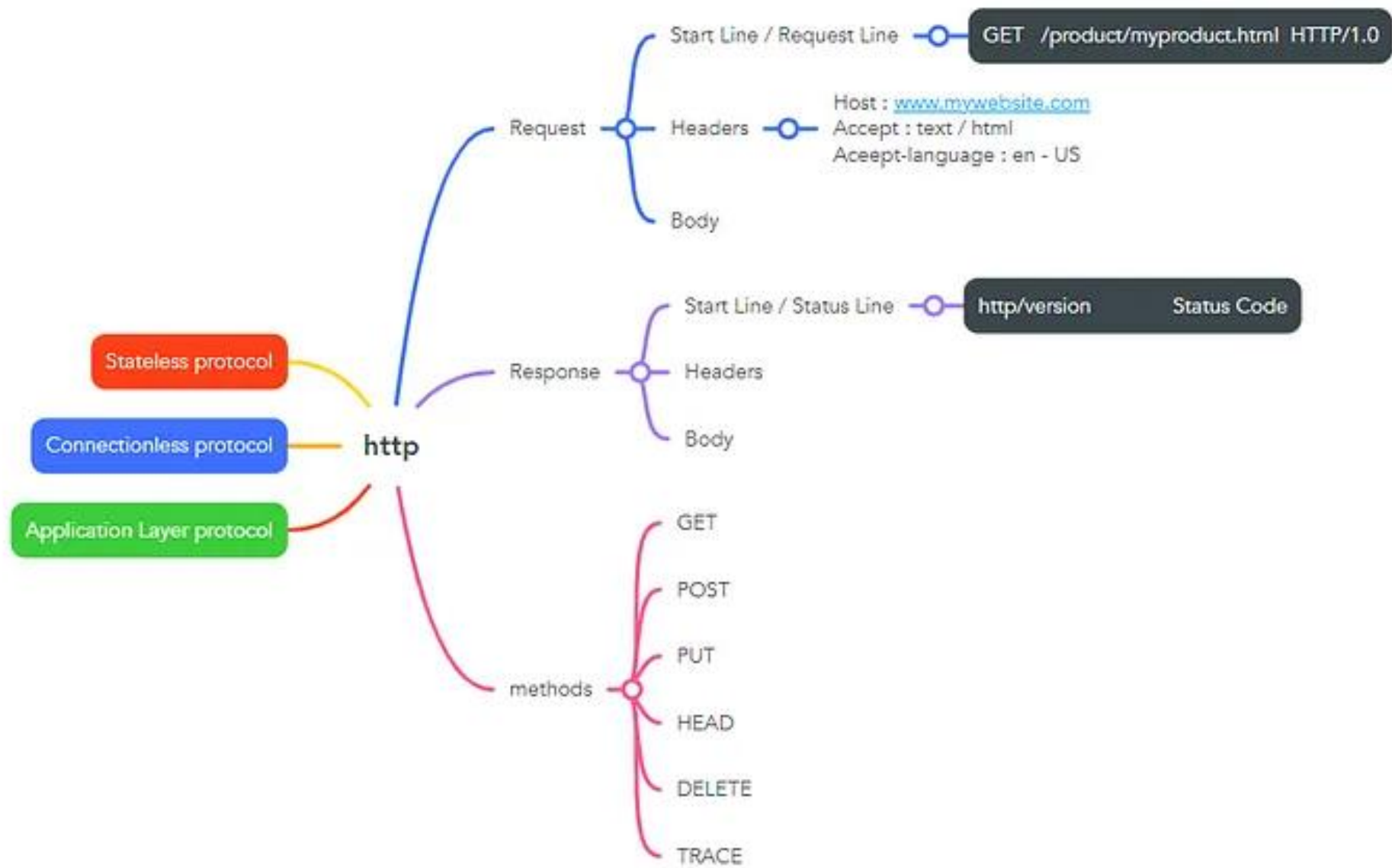
DELETE: DELETE requests to delete a resource on a server. They are used to remove resources or data that are no longer needed

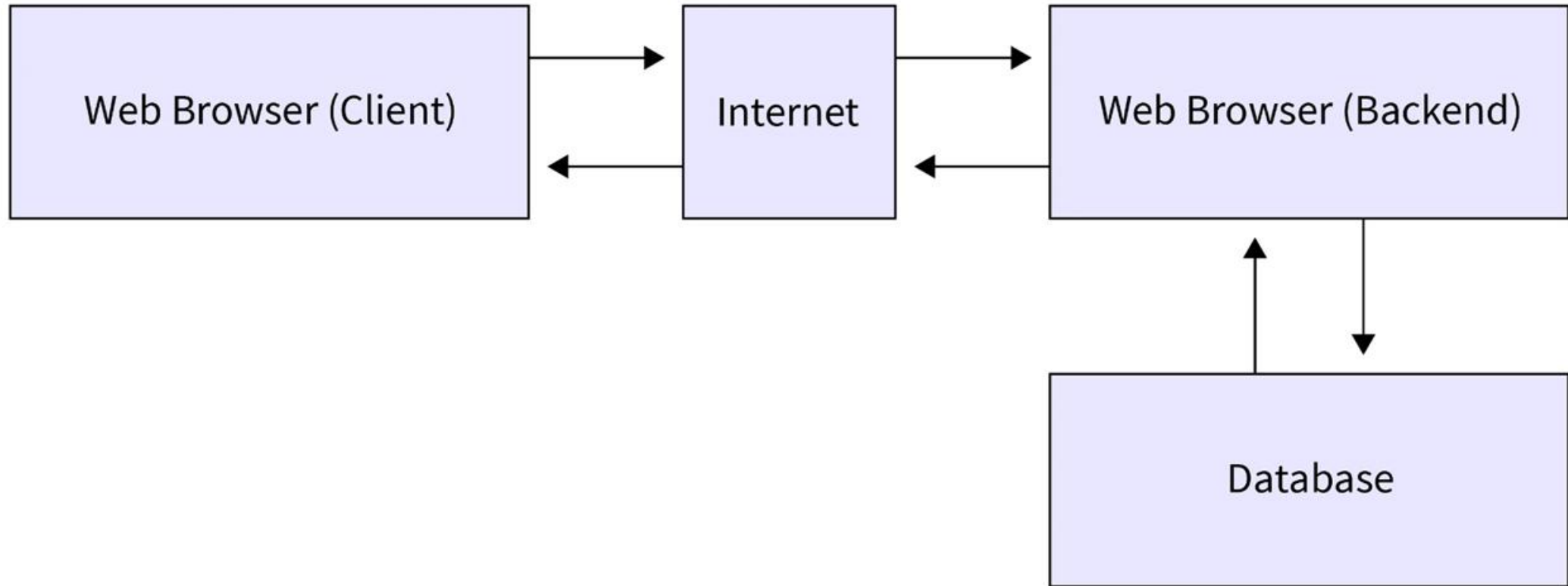
CONNECT: Connect requests are used to establish a secure connection to a server,

typically for the purpose of establishing a secure connection to a proxy server.

OPTIONS: OPTIONS requests are used to retrieve information about the communication options available for a resource.

They are typically used to retrieve information about the allowed methods and the communication options for a resource.





HTTP Connections

1. Non-Persistent
2. Persistent

Round Trip Time (RTT)

Round Trip Time is defined as the time taken by a packet of information to travel from client to server and then come back (i.e. server to client).

TCP 3-Way Handshake

1. Asking the server whether it can allow a connection to be established or not

The client sends a request to the server to open a TCP connection on which, data can be exchanged.

2. The server sends confirmation whether the connection can be established or not

After receiving the request from the client, the server sends an acknowledgment to tell the client to establish a connection.

3. Acknowledgment by the client corresponding to the response sent by the server

The client acknowledges the response (acknowledgment sent in step 2) sent by the server. After this, a connection is established between the client and the server and the data can be exchanged on this connection.

Types of HTTP Connection

Non-Persistent

In non-persistent connection HTTP, there can be at most one object that can be sent over a single TCP connection. This means that for each object that is to be sent from source to destination, a new connection will be created. HTTP/1.0 is the version of HTTP that uses a non-persistent connection.

Non-persistent HTTP is used in fetching those objects which are not needed that frequently.

Non-persistent connection HTTP requires 2 RTT (round trip time) for each object that is to be transmitted (1 RTT to open the connection and 1 RTT for transmission of data).

Persistent

In persistent connection HTTP, multiple objects can be sent over a single TCP connection. This means that multiple objects can be transmitted from source to destination on a single HTTP connection. HTTP/1.1 is the version of HTTP that uses a persistent connection.

All modern web browsers like Mozilla Firefox and Google Chrome use persistent HTTP connections.

Persistent HTTP does not require 2 RTT (round trip time) for each object that is to be transmitted. After a successful opening of the TCP connection (opening of TCP connection is done by 3-Way Handshaking which takes 1 RTT), each object will require only 1 RTT to be transmitted.

Non-Persistent Connection

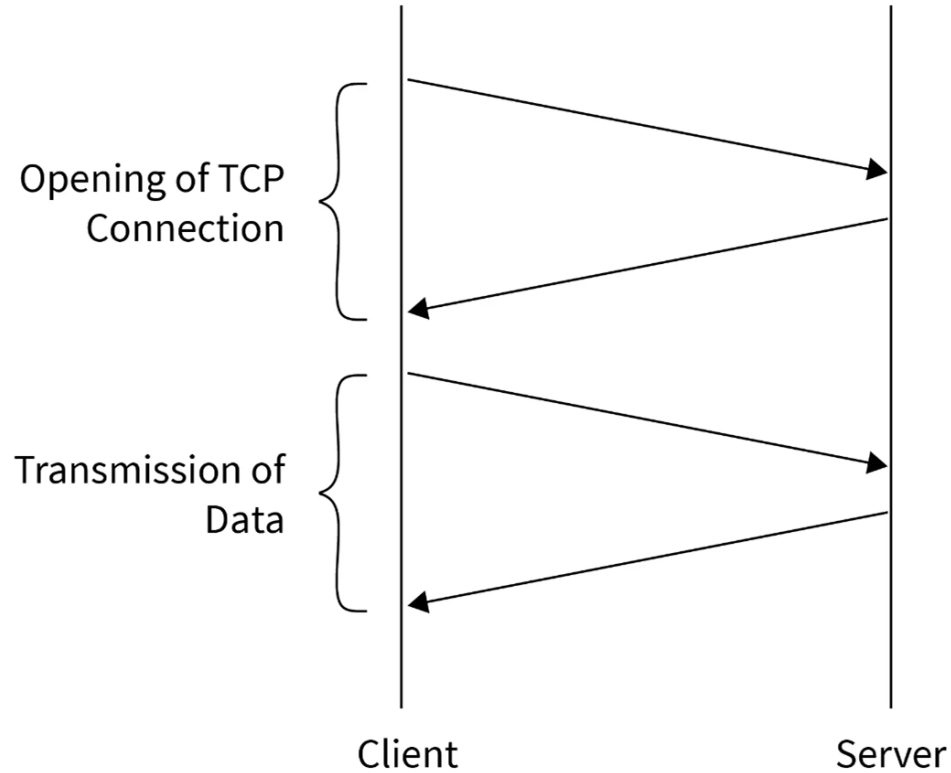
Without a Parallel Connection

In an HTTP connection without a parallel connection, each transmission takes 2 RTT (Round Trip Time) to be transmitted. The first RTT is to open a TCP connection and the second RTT corresponds to the transmission of the object on the connection.

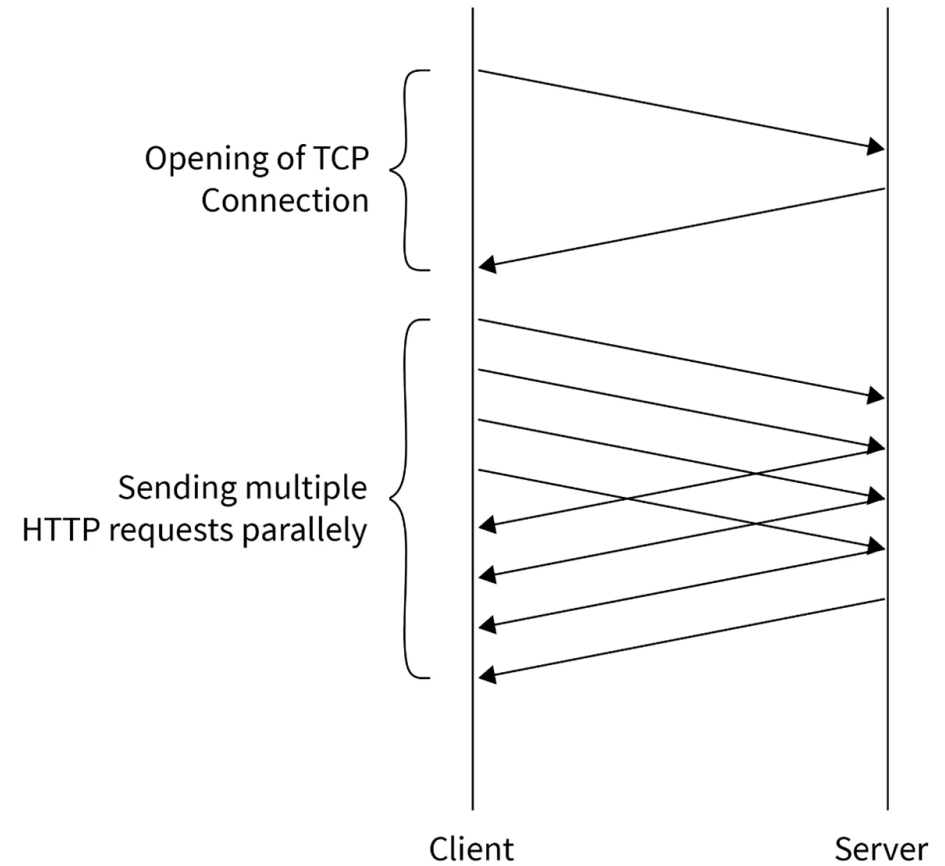
With Parallel Connection

Non-persistent HTTP needs to maintain an extra overhead to open a TCP connection each time some data needs to be transmitted over it. This makes the process of exchange of HTTP requests and responses between a client and a server slower. To make this process faster, sometimes, browsers open parallel connections in which, after the opening of a TCP connection, multiple HTTP requests are sent from client to server parallelly. This saves the time of opening a new HTTP (TCP) connection for each HTTP request. to the transmission of the object on the connection.

Non-Persistent HTTP (Without a Parallel Connection)



Non-Persistent HTTP (With Parallel Connection)



Advantages of Non-Persistent HTTP

- It does not lead to wastage of resources since the connection is opened only when some data needs to be sent over it.
- It is more secure than persistent HTTP since after sending data over the connection, the connection gets terminated and nothing can be transmitted over it once it gets terminated.

Disadvantages of Non-Persistent HTTP

- It needs to maintain an extra overhead to open a TCP connection each time some data needs to be transmitted over it.
 - It has a slow start because of the opening of a TCP connection on every data transmission.
- 
- A solid orange horizontal bar at the bottom of the slide.

Persistent Connection

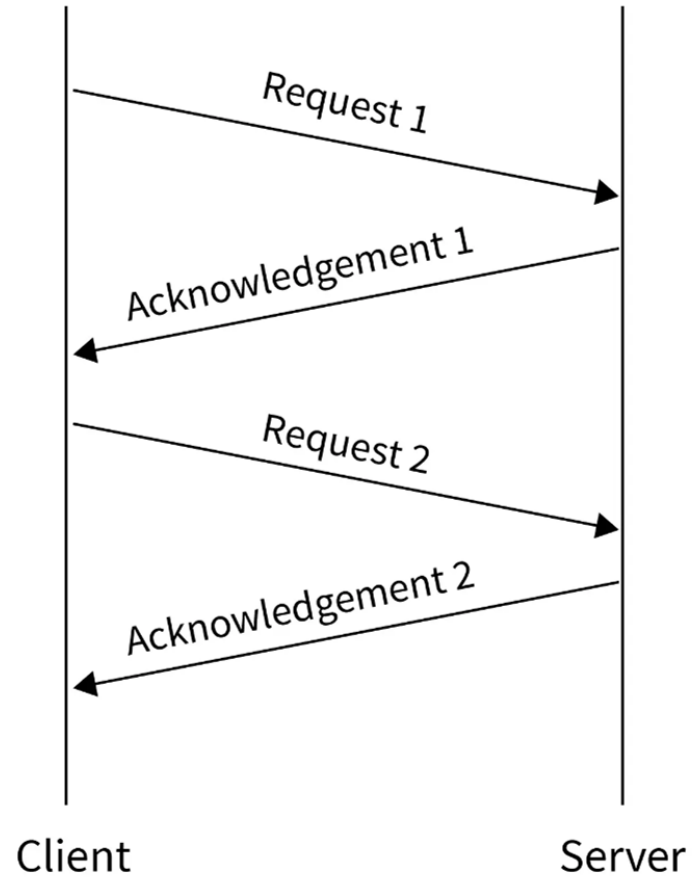
Non-Pipelined

A non-pipelined HTTP connection is a connection in which, the client can send a request to the server only if the previous request that was sent by the client has been acknowledged by the server.

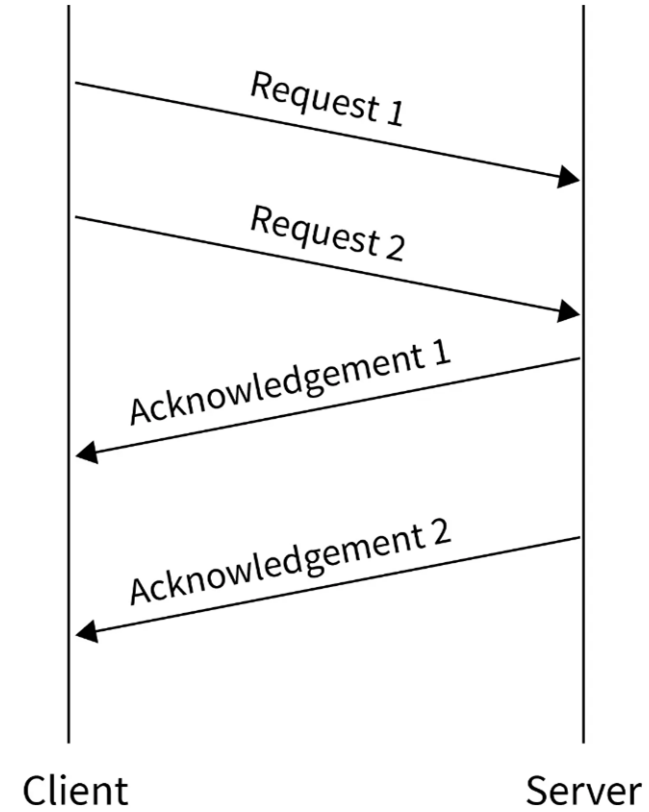
Pipelined

A pipelined HTTP connection is a connection in which, the client can send a request to the server even if the previous request that was sent by the client has not been acknowledged by the server.

Persistent HTTP (Non-Pipelined)



Persistent HTTP (Pipelined)



Advantages of Persistent HTTP

- Persistent HTTP saves CPU resources and time since the opening of the connection takes place only once.
- It gives a fast start to send any object from a source to a destination. Also, it results in relatively less network congestion and latency on subsequent HTTP requests compared to non-persistent HTTP.

Disadvantages of Persistent HTTP

- Persistent HTTP results in wastage of network resources since the connection remains open even when there is no transmission of data.
 - It is less secure than non-persistent HTTP since the connection once opened remains always open.
- 

Email protocols list

The standard email protocol list includes:

SMTP

POP3

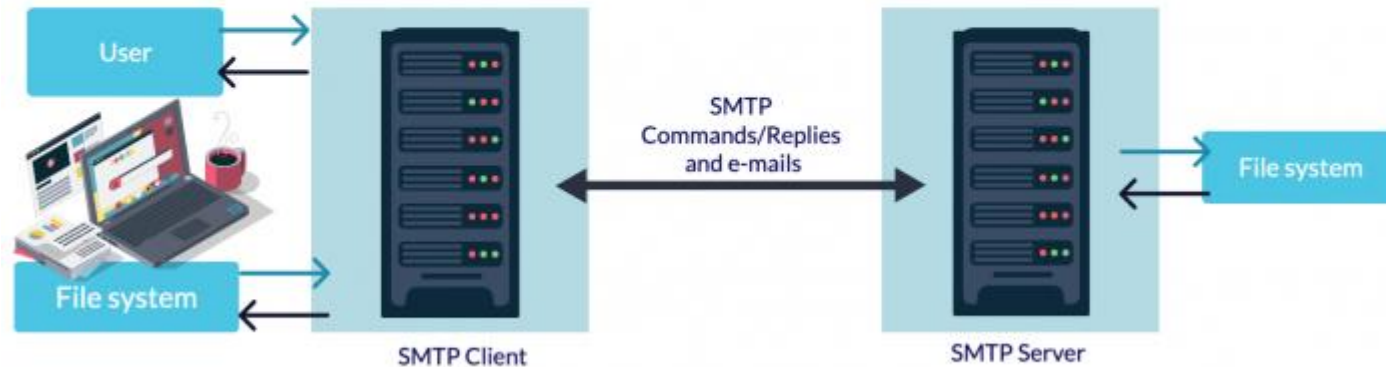
IMAP

Each of them operates differently and provides a different service for managing your email account.

What is SMTP

SMTP stands for Simple Mail Transfer Protocol, and it is responsible for sending email messages. This protocol is used by email clients and mail servers to exchange emails between computers.

How SMTP works



A mail client and the SMTP server communicate with each other over a connection established through a particular email port.

Both entities are using SMTP commands and replies to process your outgoing emails.

Simple Mail Transfer Protocol: messages can be sent from the same account on different email applications.

What is POP3?

The POP3 abbreviation stands for **Post Office Protocol version 3**, which provides access to an inbox stored in an email server. It executes the download and deletes operations for messages.

When a POP3 client connects to the mail server, it retrieves all messages from the mailbox. Then it stores them on your local computer and deletes them from the remote server.

POP3: you are able to access the messages locally in offline mode as well.

Modern POP3 clients allow you to keep a copy of your messages on the server if you explicitly select this option.

How POP3 works?



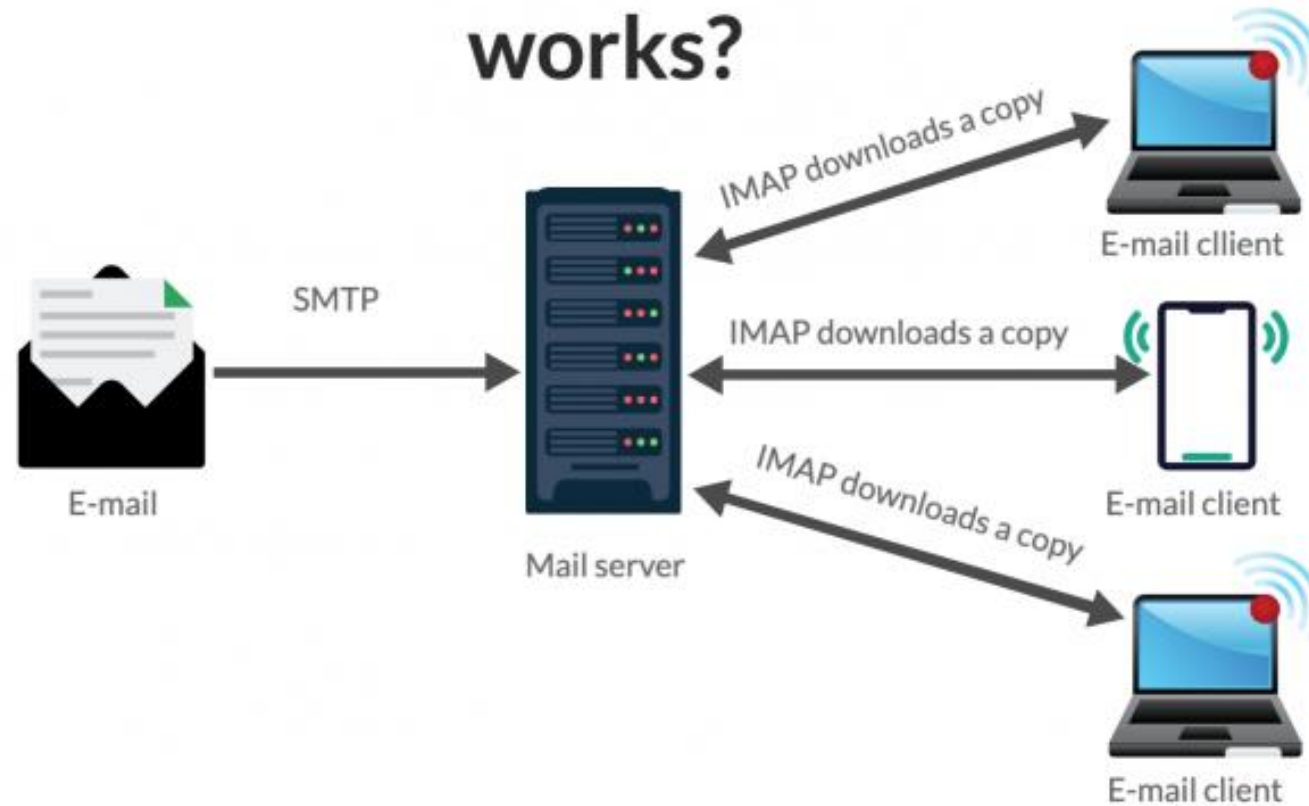
What is IMAP?

The Internet Message Access Protocol (IMAP) allows you to access and manage your email messages on the email server.

This protocol permits you to manipulate folders, permanently delete and efficiently search through messages. It also gives you the option to set or remove email flags, or fetch email attributes selectively. By default, all messages remain on the server until the user specifically deletes them.

IMAP supports the connection of multiple users to a single mail server.

How IMAP works?



Simple Mail Transfer Protocol (SMTP)

SMTP is an application layer protocol.

The client who wants to send the mail opens a TCP connection to the SMTP server and then sends the mail across the connection.

The SMTP server is always on listening mode.

As soon as it listens for a TCP connection from any client, the SMTP process initiates a connection on that port (25).

After successfully establishing the TCP connection the client process sends the mail instantly.

SMTP Protocol

The SMTP model is of two type :

End-to- end method

Store-and- forward method

The end to end model is used to communicate between different organizations whereas the store and forward method is used within an organization.

A SMTP client who wants to send the mail will contact the destination's host SMTP directly in order to send the mail to the destination.

SMTP Protocol

The SMTP server will keep the mail to itself until it is successfully copied to the receiver's SMTP.

The client SMTP is the one which initiates the session let us call it as client-

SMTP and the server SMTP is the one which responds to the session request and let us call it as receiver-SMTP.

The client- SMTP will start the session and the receiver-SMTP will respond to the request.

SMTP Protocol

In the SMTP model user deals with the **user agent** (UA) for example Microsoft outlook, netscape, Mozilla etc.

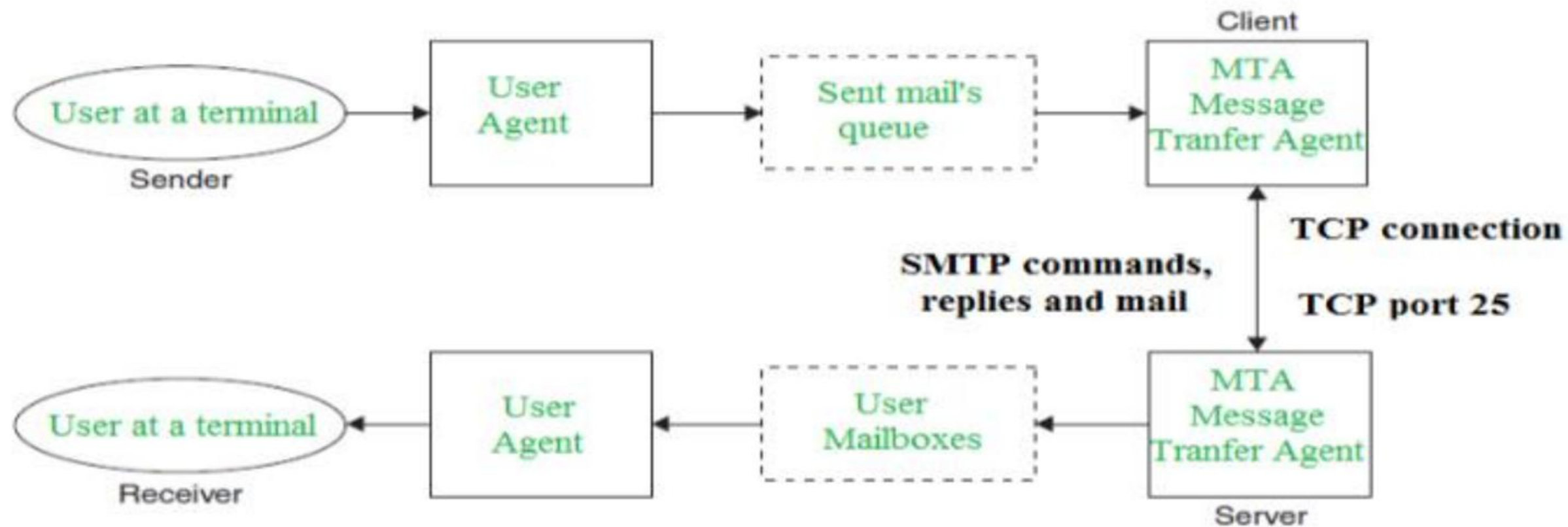
In order to exchange the mail using TCP, MTA is used.

The users sending the mail do not have to deal with the MTA

It is the responsibility of the **system admin** to set up the local MTA.

The **MTA** maintains a small queue of mails so that it can schedule repeat delivery of mail in case the receiver is not available.

The MTA delivers the mail to the mailboxes and the information can later be downloaded by the user agents.



SENDING EMAIL:

Mail is send by a series of request and response messages between the client and a server.

The message which is send across consists of a header and the body.

A null line is used to terminate the mail header.

Everything which is after the null line is considered as body of the message which is a sequence of ASCII characters.

The message body contains the actual information read by the receipt.

RECEIVING EMAIL:

The user agent at the server side checks the mailboxes at a particular time of intervals.

If any information is received it informs the user about the mail. When user tries to read the mail it displays a list of mails with a short description of each mail in the mailbox.

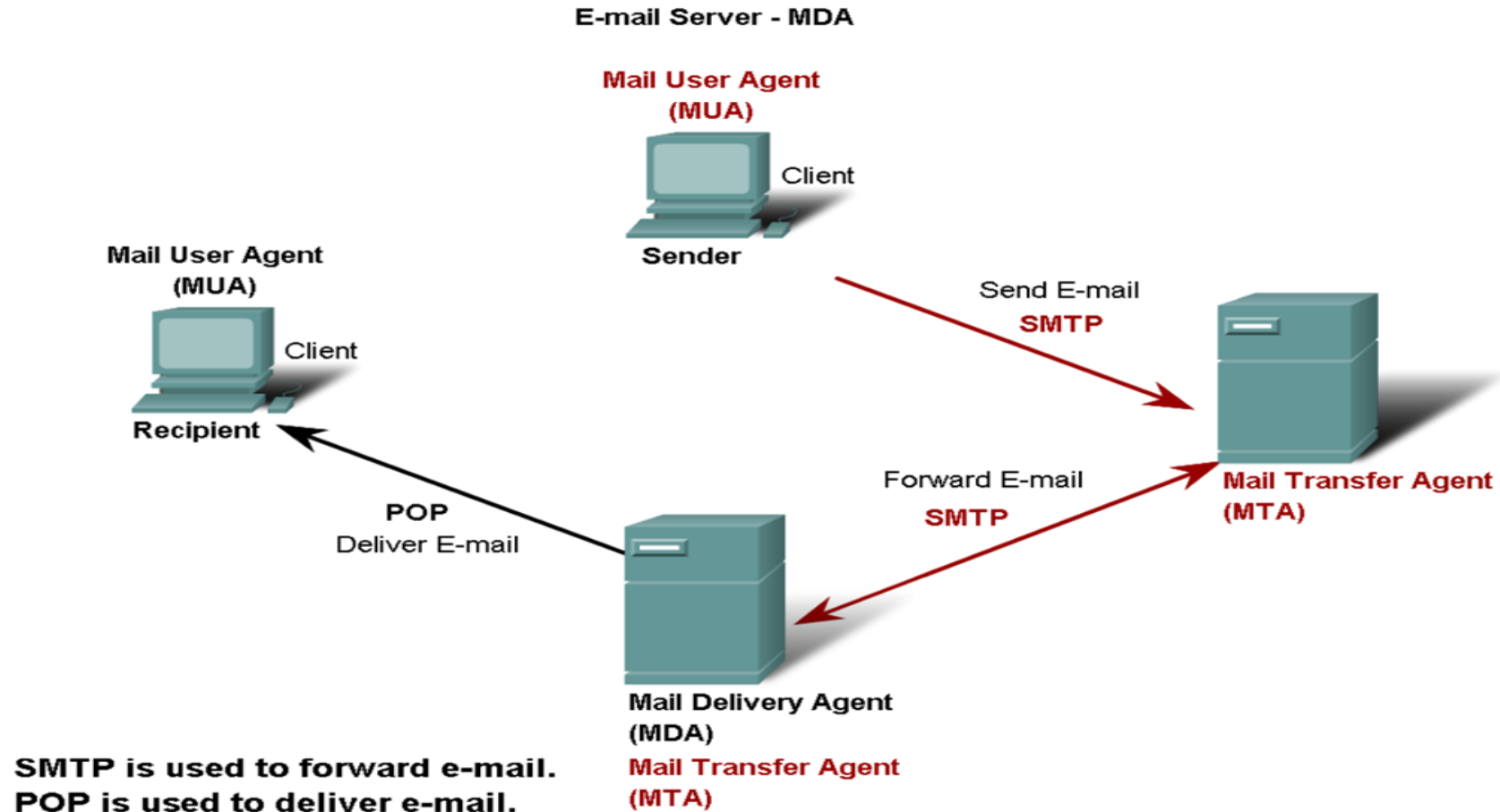
By selecting any of the mail user can view its contents on the terminal.

Main Components of SMTP

Property	Description
Body	Body text of the email.
From	Sender's email address.
FromName	Sender's name.
IsBodyHTML	Boolean value which is set to True if the body of the email message is in HTML format.
SmtpServerName	Name of server through which mail is routed.
SmtpServerPort	Port through which mail is routed.
SmtpUsername	Username for SMTP server.
SmtpPassword	Password for SMTP server.
Subject	Subject of e-mail note.
To	Email address to which you are sending the note. To send note to more than one recipient, separate addresses using commas.

Method	Description
Add Attachment	Use this method to include an attachment with an email sent using the Send method.
Send	Use this method to send an email via SMTP.

E-mail services and SMTP/POP protocols



Other Mail Protocols

SMTP - Simple Mail Transport Protocol is used on the internet, it is not a transport layer protocol but is an application layer protocol.

POP3 - Post Office Protocol version 3 is used by clients to access an internet mail server to get mail. It is not a transport layer protocol.

IMAP4 - Internet Mail Access Protocol version 4 is the replacement for POP3.

MIME - Multipurpose Internet Mail Extension is the protocol that defines the way files are attached to SMTP messages.

X.400 - International Telecommunication Union standard defines transfer protocols for sending mail between mail servers.

EMAILS ARRIVE IN SECONDS

The Complete Journey Explained



An email travels through multiple servers and protocols to reach the recipient in seconds.

1

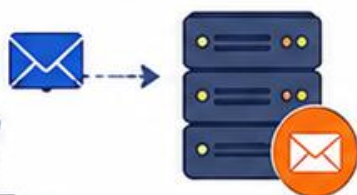
You Compose and Send



You write an email and click SEND in your email app.

2

SMTP Sends to Outgoing Mail Server



Your email client uses SMTP to send the email to the outgoing mail server.

3

DNS Finds the Recipient's Mail Server



The outgoing server asks DNS for the recipient domain's MX record to find the correct mail server.

4

SMTP Transfers the Email



The email is transferred from the outgoing server to the recipient's mail server using SMTP.

5

Email Stored in Recipient Mail Server



The recipient's mail server stores the email securely in the recipient's mailbox (inbox/spam, etc.).

6

Recipient Retrieves the Email (IMAP/POP3)



The recipient's email app uses IMAP or POP3 to retrieve or sync the email from the server.

7

You Read the Email



The email appears in the recipient's inbox and is ready to read.

Key Protocols Involved

Stage	Protocol / Tech	Purpose
1 2	SMTP	Sends or submits email from client to server
3	DNS	Finds the mail server for the recipient's domain
3	MX Record	Specifies the mail server responsible for domain
4	SMTP	Transfers email between mail servers
6	IMAP	Synchronizes email across devices (keeps on server)
6	POP3	Downloads email to device (may remove from server)

What is MX Record?

MX (Mail Exchange) record specifies the mail server responsible for accepting emails for a domain.

Through the Internet



Email may pass through multiple routers and servers.

Quick Facts

- ✓ DNS does NOT carry the email. It only tells where to send.
- ✓ SMTP is used for sending and relaying emails.
- ✓ IMAP keeps emails on the server and syncs across devices.
- ✓ That's why emails arrive in seconds!



THE COMPLETE JOURNEY – AT A GLANCE



Thanks to standard protocols and a global network of servers, your email travels securely and reaches the recipient within seconds!



FTP

FILE TRANSFER PROTOCOL

File Transfer Protocol (FTP)

File Transfer Protocol (FTP) is a standard Internet protocol for transmitting files between computers on the Internet over TCP/IP connections.

FTP is a client-server protocol that relies on two communications channels between client and server: a command channel for controlling the conversation and a data channel for transmitting file content. Clients initiate conversations with servers by requesting to download a file. Using FTP, a client can upload, download, delete, rename, move and copy files on a server. A user typically needs to log on to the FTP server, although some servers make some or all of their content available without login, also known as anonymous FTP.

FTP Data Types

FTP data types define how file contents are represented and transmitted between the client and server during file transfer.

ASCII: This data type is used for transferring text files with character encoding based on the ASCII standard.

EBCDIC: This data type is used for transferring text files encoded in IBM's Extended Binary Coded Decimal Interchange Code.

Image (Binary): This data type transfers files as a continuous stream of bytes without any modification.

Local: This data type is used for transferring files that contain data stored in logical bytes of a specified bit length.

Common FTP Commands

An FTP client is a software application that enables a user to connect to an FTP server, authenticate, and perform file transfer operations over a network using the client–server model.

get filename – Downloads a single file from the server.

mget filename – Downloads multiple files from the server.

ls – Displays the list of files in the current server directory.

FTP Replies

- 200** – Command executed successfully.
- 221** – Service closing control connection.
- 331** – Username accepted; password required.
- 502** – Command not implemented.
- 503** – Incorrect sequence of commands.
- 504** – Command not supported for that parameter.
- 530** – User not logged in (authentication failed).
- 551** – Requested action aborted: page type unknown.

Types of FTP

1. Anonymous FTP

It is a file transfer method that allows users to access publicly available files on a server without requiring a personal user account.

Users log in using the username “anonymous”.

Usually provides read-only access.

Does not allow modification of server directories.

Commonly used for distributing public software and documents.

2. Password-Protected FTP

This is a secure access method that requires valid user credentials to access and manage files on an FTP server.

Requires username and password authentication.

Access permissions are controlled by the administrator.

Supports file upload, download, and modification.

Suitable for private or organizational file sharing.

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3. FTPS (FTP Secure)

FTPS is an extension of FTP that adds SSL/TLS encryption to secure data transmission between client and server.

Encrypts login credentials and transfers data.

Operates over TCP (commonly port 21).

Supports implicit and explicit encryption modes.

Maintains compatibility with standard FTP commands.



4. SFTP (SSH File Transfer Protocol)

This is a secure file transfer protocol that operates over the SSH protocol and provides encrypted communication for file management.

Uses port 22 for communication.

Encrypts both authentication and file data.

Uses a single secure connection (no separate data channel).

Provides secure remote file access and management.

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Benefits of File Transfer Protocol

- You can transfer multiple files and directories simultaneously, rather than having to transfer them one at a time.
- You can resume transfers if you lose your connection.
- FTP supports queueing, which means that users can add files to a transfer queue. The files in this queue will be transferred in the future.
- There's no limit on the number of files or the maximum size of the files that you can transfer.
- FTP supports scheduled transfers.
- FTP allows you to abort ongoing transfers.

Security Issues

- Transmits data and credentials in plain text, making interception easy.
- Vulnerable to attacks such as sniffing, spoofing, brute force, and port scanning.
- Does not provide built-in encryption for secure communication.
- If intercepted, attackers can read or modify transferred data.

TCP Port No

TCP/IP Well Known Port Numbers (0 to 1023)

Port #	Portocol	Description
20	TCP	FTP - data port (FTP-d)
21	TCP	FTP - control (command) port (FTP-c)
22	TCP, UDP	SSH (Secure Shell) - used for secure logins, file transfers (scp, sftp) and port forwarding
23	TCP, UDP	Telnet protocol - unencrypted text communications

Telnet

Telnet is a protocol that enables you to connect to remote computers and local computers over a TCP/IP network, over TCP port 23.

By default, **Telnet** is disabled in recent **Windows** environments.

To enable **Telnet command** line utilities: Click Start > Control Panel.

Control Functions

Interrupt Process (IP)

- suspend/abort process.

Abort Output (AO)

- send no more output to user's terminal.

Are You There (AYT)

- check to see if system is still running.

Erase Character (EC)

- delete last character sent

Erase Line (EL)

- delete all input in current line.

DHCP

Seamless network connectivity is achieved by automatically assigning IP addresses and configuration settings to devices using the Dynamic Host Configuration Protocol (DHCP).

- Assigns a unique IP address to each device.
- Provides the subnet mask to define the local network range.
- Configures the default gateway for communication outside the local network.
- Supplies DNS server addresses for domain name resolution.

Components

DHCP Server: The device (router/server) that manages the DHCP service, keeps the IP pool, and assigns IP addresses + configuration to clients.

DHCP Relay: A router/switch feature that forwards DHCP requests and replies between clients and the server when they are in different networks (different subnets).

DHCP Client: Any device (PC, mobile, printer, IoT) that requests an IP address and settings from the DHCP server.

IP Address Pool: The range of IP addresses available for allocation to clients (often with exclusions/reservations).

Lease: The time period for which an assigned IP address is valid; the client must renew the lease to keep using it.

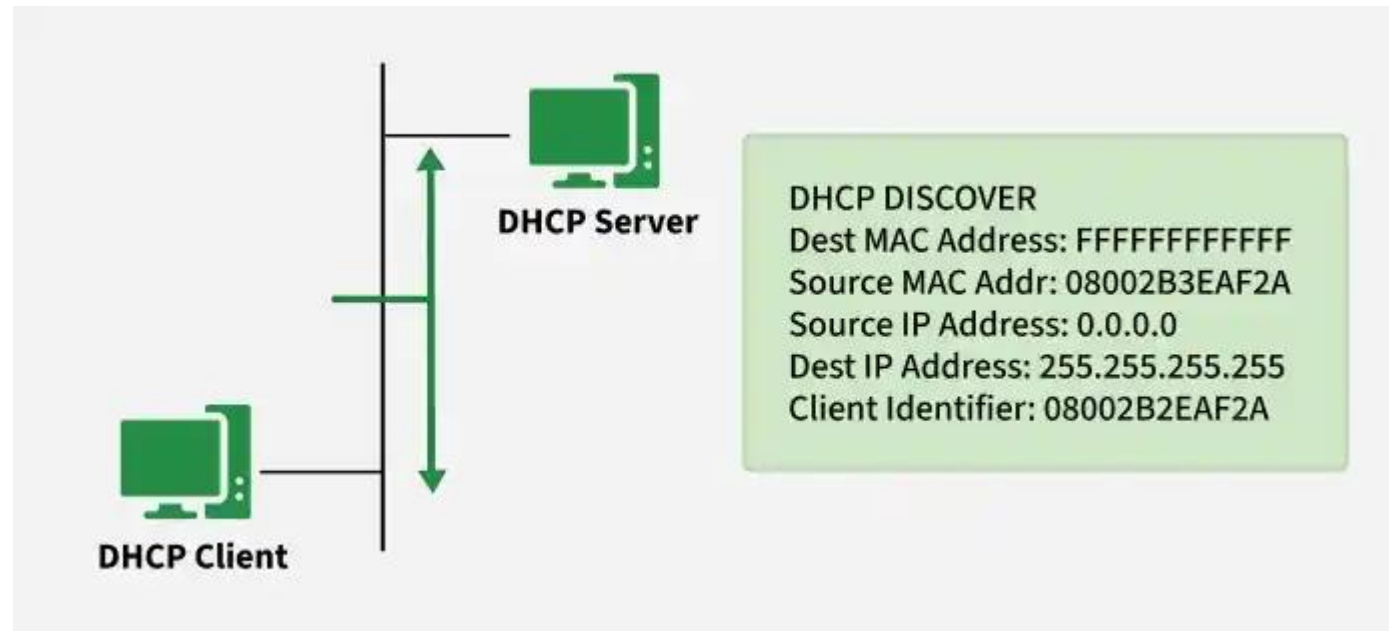
DHCP Options: Additional configuration parameters provided by the DHCP server along with the IP address, such as DNS servers, domain name, NTP servers, and other network-specific settings.

Default Gateway: The router IP given to the client so it can communicate outside the local network (reach the internet/other networks).

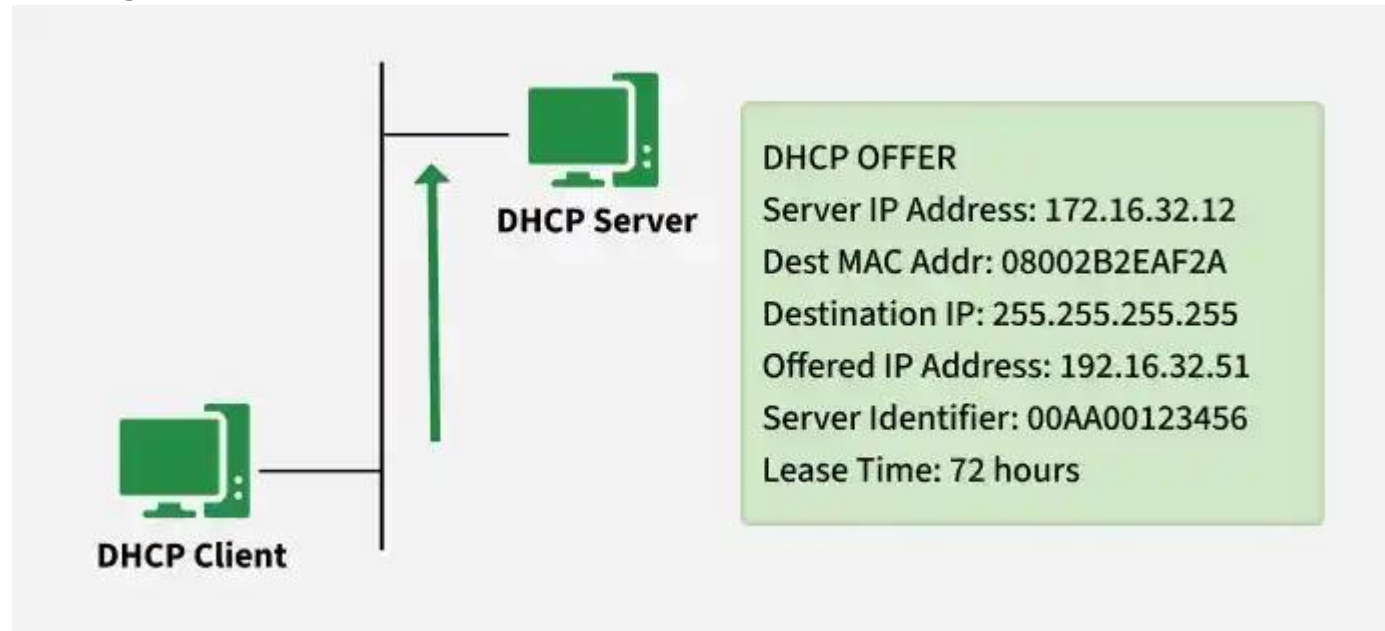
DNS Servers: DNS addresses provided to the client so it can convert domain names (like google.com) into IP addresses.

Working

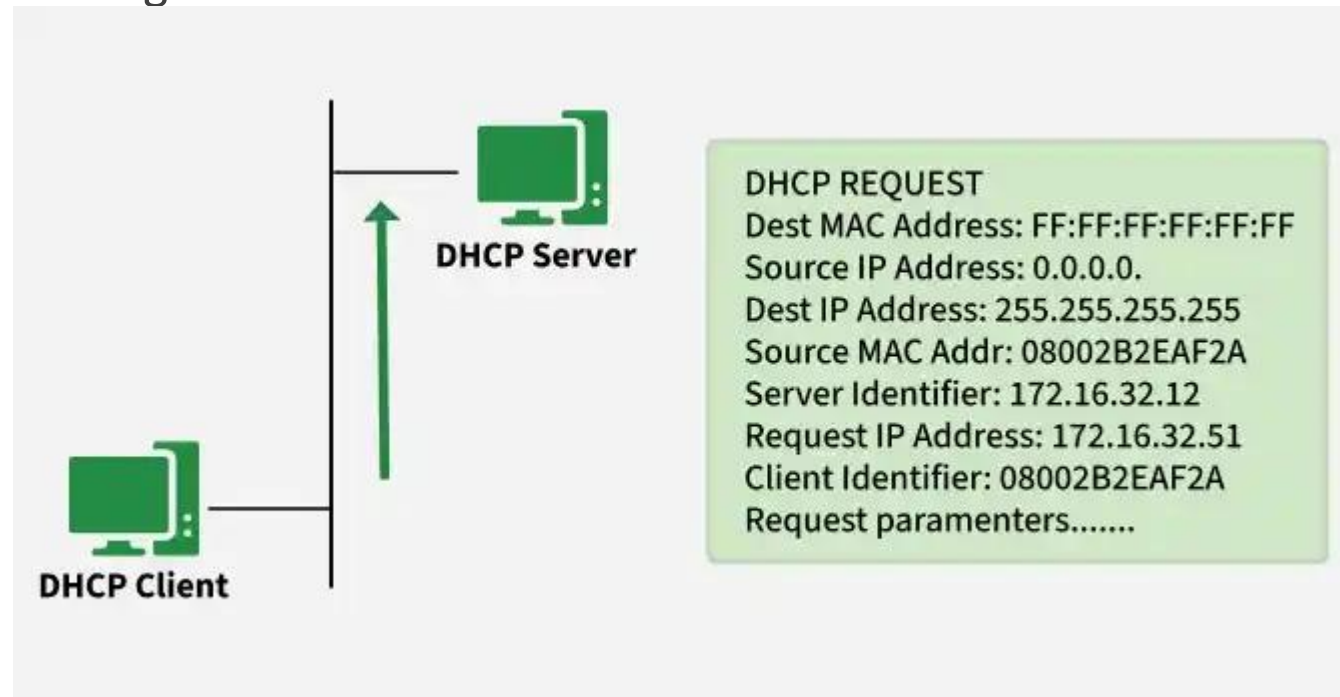
1. DHCP Discover Message



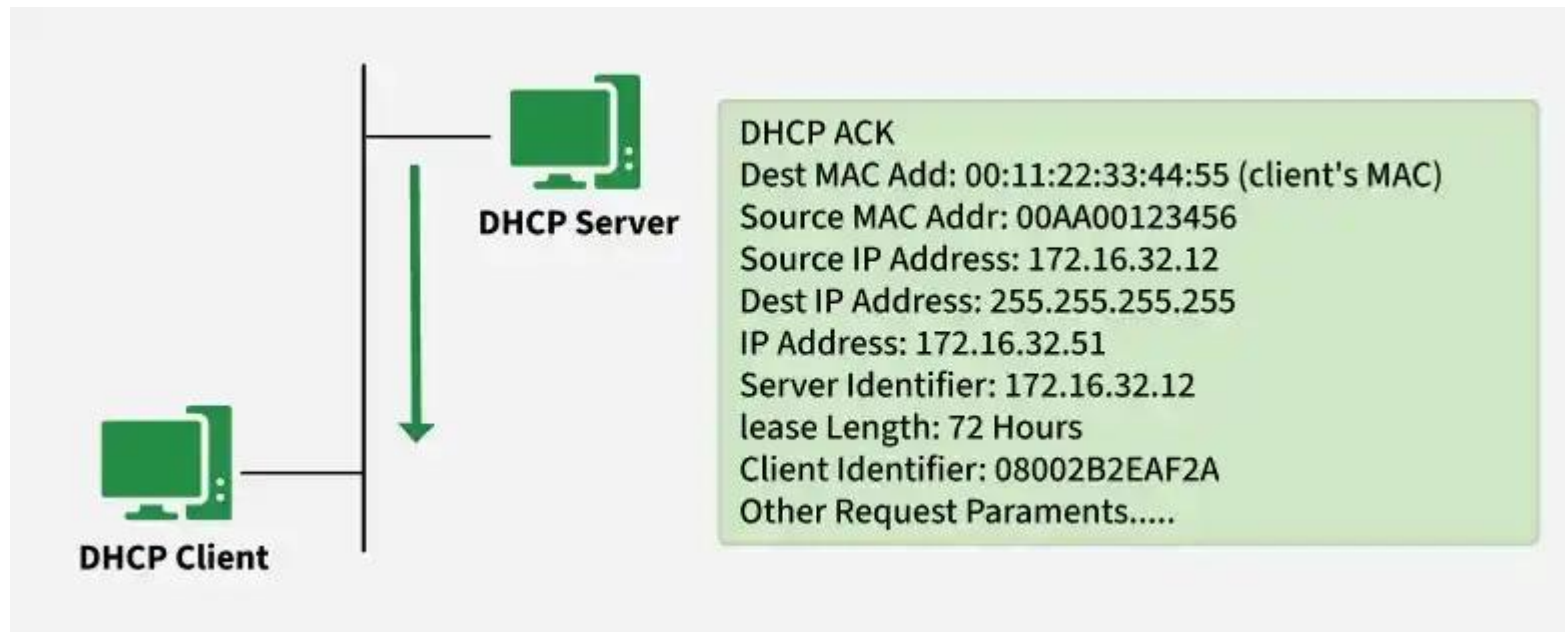
2. DHCP Offer Message



3. DHCP Request Message



4. DHCP Acknowledgment Message



5. DHCP Negative Acknowledgment Message

DHCP NAK is sent by the server to notify the client that the requested IP configuration is invalid and cannot be assigned.

Sent by: DHCP server

Purpose: Reject the client's DHCP Request

Requested IP is outside the configured scope

IP is already allocated to another device

Client has moved to a different network

Client must restart the DHCP process from Discover

6. DHCP Decline

DHCP Decline is sent by the client when it detects that the offered IP address is already in use on the network.

Sent by: DHCP client

Purpose: Inform the server that the offered IP is not safe to use

Typical trigger: Client detects an IP conflict using an ARP probe / gratuitous ARP

Result: Server marks the IP as unavailable and selects another address for future allocation.



7. DHCP Release

DHCP Release allows a client to return its assigned IP address to the server before the lease expires.

Sent by: DHCP client

Purpose: Free the IP for reuse

Occurs when a device disconnects, shuts down, or disables the interface

Helps maintain efficient IP address utilization



8. DHCP Inform

DHCP Inform is used when a client already has a manually configured (static) IP but needs additional network settings from the DHCP server.

Sent by: DHCP client

Purpose: Request configuration parameters like DNS, domain name, NTP, etc., without obtaining a new IP.

Server response: Sends a DHCP ACK containing only configuration options.

Delivery: Typically unicast, since the client already has a valid IP.

Security Concerns with DHCP

DHCP has no built-in authentication, so attackers on the same LAN can misuse it to disrupt service or redirect traffic.

DHCP Starvation (IP Pool Exhaustion): Floods DHCP requests with spoofed MACs to consume all available leases, blocking legitimate clients.

Rogue DHCP Server: A fake server gives clients incorrect IP settings (gateway/DNS), taking control of their network path.

Man-in-the-Middle: Done by pushing attacker-controlled gateway/DNS via rogue DHCP to intercept or alter traffic.

DNS Misuse: Malicious DHCP options can point clients to unauthorized DNS servers for redirection/phishing.

Protection Against DHCP Attacks

Enable DHCP Snooping on switches and allow DHCP replies only on trusted ports.

Apply Port Security or rate-limiting to reduce spoofed requests and prevent starvation.

Use IP/MAC filtering + monitoring/log analysis to detect and block abnormal lease activity.